

Gün Kaynar

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Research Interests

My research develops methods across machine learning, AI for science, and computational biology. In **machine learning**, I work on reasoning distillation for diffusion language models, study failure modes and reliability of tool-using LLM agents, and develop techniques for deployment-time shortcut mitigation and imbalanced classification. In collaboration with Arm, I develop simulation frameworks, orchestration and scheduling systems, error handling, and agentic supervision for **programmable cloud laboratories**, enabling AI agents to autonomously operate instruments. Separately, I'm part of Anthropic's Model Hardware Standard (MHS) research preview, letting AI agents safely operate lab equipment and run experiments autonomously. I study **reinforcement learning** for multi-objective sequence optimization problems in biology, and build structure-aware models that integrate biophysical simulation with machine learning for modeling translation. In **mRNA therapeutics**, computationally optimized coding sequences from our design methods undergo experimental production and validation. I also work on **clinical informatics**, including adverse event prediction from electronic health records, differentially private synthetic EHR generation, and disease monitoring in transplant patients.

Education

- Aug 2024 **Carnegie Mellon University, Pittsburgh, PA, Ph.D. student, School of Computer Science, Computational Biology, Advisor:** Carl Kingsford.
Present
Sep 2021 **Bilkent University, Ankara, Turkey, M.Sc., Computer Science, Advisor:** A. Ercument
Aug 2024 Cicek.
Thesis: Pathway-Informed Deep Learning Model for Survival Analysis and Pathological Classification of Gliomas
Sep 2015 **Bogazici University, Istanbul, Turkey, B.Sc., Molecular Biology and Genetics.**
Aug 2021
Feb 2020 **Universitat de Barcelona, Barcelona, Spain, Exchange student.**
Jul 2020

Honors and Awards

- 2026 **Graduate Student Assembly/Provost Conference Funds, Carnegie Mellon University,** Travel funding to present at RECOMB 2026.
2026 **CBD Travel Award, Ray and Stephanie Lane Computational Biology Department, Carnegie Mellon University,** Travel funding to present at ISMB 2026.
2026 **ISMB 2026 Conference Fellowship.**
2026 **Randy Bryant Endowed Fellowship Fund, School of Computer Science, Carnegie Mellon University,** Recognizes potential for sustained academic excellence and groundbreaking research. Provides partial funding for tuition, stipend, and support costs..
2025 **HPC and Data Science Summer Institute Scholarship, San Diego Supercomputer Center.**

- 2024 **Featured Article in Nature Communications Editors' Highlights**, *50 Best Papers in Biotechnology and Methods category of Nature Communications*.
- 2024 **Carnegie Mellon University SCS Ph.D. Research Assistantship**, full tuition waiver, health insurance and stipend during Ph.D. studies..
- 2024 **Tri-Institutional Ph.D. Fellowship**, full tuition waiver, health insurance and stipend during Ph.D. studies..
- 2024 **UCLA Ph.D. Fellowship**, full tuition waiver, health insurance and stipend during Ph.D. studies..
- 2022 **International Scientific Publication Incentive Program**, Scientific and Technological Research Council of Turkey (TÜBİTAK).
- 2021-2024 **Bilkent University Comprehensive Scholarship**, full tuition waiver, stipend, health insurance, and accommodation during M.Sc.
- 2021-2024 **Scientist Supporting Program Scholarship**, Scientific and Technological Research Council of Turkey, Merit-based monthly stipend during M.Sc.
- 2020 **Erasmus Mobility Grant**, European Commission, full coverage for mobility.

Selected Press

- **Previewing the Model Hardware Standard**, Anthropic, August 2026. Featured as part of the CMU research team in Anthropic's announcement of the Model Hardware Standard (MHS) research preview, demonstrating AI-agent orchestration of laboratory instruments for autonomous dose-response experiments. Also covered by CMU School of Computer Science on X (Twitter) and LinkedIn.
- **Center for Innovation in Health**, Carnegie Mellon University School of Computer Science. Featured researcher for work on augmenting electronic health records for adverse event detection (TASER-AE).

Publications (Google Scholar, ResearchGate)

Journal Articles

- **Gün Kaynar**, Carl Kingsford, "seq2ribo: Structure-aware integration of machine learning and simulation to predict ribosome location profiles from RNA sequences", in *Bioinformatics*, June 2026. [Online].
- Mehmet Alper Yilmaz*, Ahmet Arda Ceylan*, **Gün Kaynar***, A Ercument Cicek, "LYCEUM: Learning to call copy number variants on low coverage ancient genomes", in *Bioinformatics*, July 2025. [Online]. Available: https://academic.oup.com/bioinformatics/article/41/Supplement_1/i285/8199375
- *: equal contribution
- Berk Mandiracioglu, Furkan Ozden, **Gün Kaynar**, Mehmet Alper Yilmaz, Can Alkan, A. Ercument Cicek, "ECOLE: Learning to call copy number variants on whole exome sequencing data", in *Nature Communications*, Jan 2024. [Online]. Available: <https://www.nature.com/articles/s41467-023-44116-y>
- **Gün Kaynar**, Doruk Cakmakci, Caroline Bund, Julien Todeschi, Izzie Jacques Namer, A Ercument Cicek, "PiDeeL: Metabolic Pathway-Informed Deep Learning Model for Survival Analysis and Pathological Classification of Gliomas", in *Bioinformatics*, November 2023. [Online]. Available: <https://academic.oup.com/bioinformatics/article/39/11/btad684/7413171>
- Doruk Cakmakci*, **Gün Kaynar***, Caroline Bund, Martial Piotto, Francois Proust, Izzie Jacques Namer, A Ercument Cicek, "Targeted metabolomics analyses for brain tumor margin assessment during surgery", in *Bioinformatics*, June 2022. [Online]. Available: <https://academic.oup.com/bioinformatics/article/38/12/3238/6581075>
- *: equal contribution

Peer-Reviewed Conference Proceedings/Talks

- Jiayi Li, Shijie Tang, **Gün Kaynar**, Shiyi Du, Carl Kingsford, “Models Know Their Shortcuts: Deployment-Time Shortcut Mitigation”, in *AACL-IJCNLP 2026*, Hengqin, China, November 2026.
- **Gün Kaynar**, Carl Kingsford, “seq2ribo: Structure-aware integration of machine learning and simulation to predict ribosome location profiles from RNA sequences”, in *ISMB 2026*, Washington, DC, US, July 2026.
- Sina Barazandeh, **Gün Kaynar**, Arth Banka, Joshua Kangas, Carl Kingsford, Jose Lugo-Martinez, “Agentic Fault Recovery for Autonomous Orchestration of Programmable Cloud Laboratories”, in *AI Scientist Summer Workshop*, Microsoft Research New England, Cambridge, MA, US, August 2026.
- Mehmet Alper Yilmaz*, Ahmet Arda Ceylan*, **Gün Kaynar***, A Ercument Cicek, "LYCEUM: Learning to call copy number variants on low coverage ancient genomes", in *ISMB/ECCB 2025*, Liverpool, UK, July 2025.

*: equal contribution

- Berk Mandiracioglu, Furkan Ozden, **Gün Kaynar**, Mehmet Alper Yilmaz, Can Alkan, A. Ercument Cicek, "ECOLE: Learning to call copy number variants on whole exome sequencing data", in *RECOMB 24*, Cambridge, MA, US, April-May 2024.
- Berk Mandiracioglu, Furkan Ozden, **Gün Kaynar**, Mehmet Alper Yilmaz, Can Alkan, A. Ercument Cicek, "ECOLE: Learning to call copy number variants on whole exome sequencing data", in *HIBIT 23*, Ankara, Turkey, October 2023.
- **Gün Kaynar**, Doruk Cakmakci, Caroline Bund, Julien Todeschi, Izzie Jacques Namer, A Ercument Cicek, "Pathway-Informed Deep Learning Model for Survival Analysis and Pathological Classification of Gliomas", in *RECOMB CCB*, Istanbul, Turkey, April 2023.

Preprints / Under Review

- **Gün Kaynar**, Boshen Yan, Jiayi Li, Shiyi Du, Carl Kingsford, “When Tools Fail Quietly, LLM Agents State Wrong Answers as Fact”, under review in *ICLR 2027*.
- **Gün Kaynar**, Carl Kingsford, “When Reasoning Traces Overrun the Canvas of Diffusion LLMs”, under review in *ICLR 2027*.
- **Gün Kaynar**, Jiayi Li, Yuxin Lu, Ahhyun Lee, Shiyi Du, Guillaume Marçais, and Carl Kingsford, “Practical multi-objective optimization of mRNA coding sequences using reinforcement learning and hybrid decoding”, manuscript in preparation.
- Sina Barazandeh*, **Gün Kaynar***, Arth Banka, Joshua Kangas, Carl Kingsford[†], Jose Lugo-Martinez[†], “simPCL: An Open-Framework Simulator and Benchmark Generator for Programmable Cloud Laboratories”, manuscript in preparation.

*: equal contribution, [†]: co-senior authors

- Arth Banka, Sina Barazandeh, **Gün Kaynar**, Joshua Kangas, Carl Kingsford, Jose Lugo-Martinez, “Disturbance-Aware Rescheduling, Backtracking, and Cost-Aware Planning for Time-Constrained Laboratory Automation Scheduling”, manuscript in preparation.
- Shiyi Du, **Gün Kaynar**, Jiayi Li, Zhaoyi You, Shijie Tang, Carl Kingsford, “CodonRL: Multi-Objective Codon Sequence Optimization Using Demonstration-Guided Reinforcement Learning”, under review in *Nature Communications*, 2026.
- **Gün Kaynar**, Zhaoyi You, Carl Kingsford, “TIGER: Targeted Iterative Generation and Refinement for Imbalanced Classification”, manuscript in preparation.
- **Gün Kaynar**, Zhaoyi You, Richard D. Boyce, Takahiro Yakoh, Carl Kingsford, “Augmenting Electronic Health Records for Adverse Event Detection”, in revision in *Journal of Biomedical Informatics*, 2026.

- **Gün Kaynar**, Chethan Ashokkumar, Rakesh Sindhi, et al., “Viremia and PTLTD detection method for EBV in pediatric liver transplants”, manuscript in preparation.
- Aadyaa Maddi, **Gün Kaynar**, et al., “Differentially Private Generation of Temporal Electronic Health Record Data”, manuscript in preparation.

Posters

- **Gün Kaynar**, Boshen Yan, Carl Kingsford, “When Tools Fail, Agents Fabricate: Silent Failure and Verification-First Oversight in Tool-Using LLM Agents”, in *NeurIPS 2026 Workshop Who Verifies the Agents? Toward Reliable Agent Development*, Sydney, Australia, December 2026.
- **Gün Kaynar**, Carl Kingsford, “seq2ribo: Structure-aware integration of machine learning and simulation to predict ribosome location profiles from RNA sequences”, in *CPCB 20th Anniversary Research Symposium*, Pittsburgh, PA, US, August 2026.
- Sina Barazandeh*, **Gün Kaynar***, Arth Banka, Joshua Kangas, Carl Kingsford[†], Jose Lugo-Martinez[†], “simPCL: An Open-Framework Simulator and Benchmark Generator for Programmable Cloud Laboratories”, in *CPCB 20th Anniversary Research Symposium*, Pittsburgh, PA, US, August 2026.

*: equal contribution, [†]: co-senior authors

- Sina Barazandeh, **Gün Kaynar**, Arth Banka, Joshua Kangas, Carl Kingsford, Jose Lugo-Martinez, “Accurate Simulation and LLM-Enabled Orchestration for Fault-Tolerant Programmable Cloud Laboratories”, in *AI Scientist Summer Workshop*, Microsoft Research New England, Cambridge, MA, US, August 2026.
- **Gün Kaynar**, Carl Kingsford, “seq2ribo: Structure-aware integration of machine learning and simulation to predict ribosome location profiles from RNA sequences”, in *RECOMB 2026*, Thessaloniki, Greece, May 2026.
- Sina Barazandeh*, **Gün Kaynar***, Arth Banka, Joshua Kangas, Carl Kingsford[†], Jose Lugo-Martinez[†], “An Open-Framework Simulator and Benchmark Generator for Programmable Cloud Laboratories”, in *SciFM 2026: Foundation Models and AI Agents for Science*, University of Chicago, Chicago, IL, US, May 2026.

*: equal contribution, [†]: co-senior authors

- **Gün Kaynar**, Carl Kingsford, "Modeling Ribosome Dynamics with Structure-Aware TASEP", in *Annual CPCB program retreat*, Pittsburgh, PA, US, August 2025.
- **Gün Kaynar**, Doruk Cakmakci, Caroline Bund, Julien Todeschi, Izzie Jacques Namer, A Ercument Cicek, "PiDeeL: Pathway-Informed Deep Learning Model for Survival Analysis and Pathological Classification of Gliomas", in *16th The International Symposium on Health Informatics and Bioinformatics*, Ankara, Turkey, October 2023.
- **Doruk Cakmakci***, **Gün Kaynar***, Caroline Bund, Martial Piotto, Francois Proust, Izzie Jacques Namer, A Ercument Cicek, "Targeted metabolomics analyses for brain tumor margin assessment during surgery", in *RECOMB 2023*, Istanbul, Turkey, April 2023.

*: equal contribution

Patents

- **Gün Kaynar**, Carl Kingsford, “seq2ribo: Structure-aware integration of machine learning and simulation to predict ribosome location profiles from RNA sequences”, patent application filed.
- Shiyi Du, **Gün Kaynar**, Jiayi Li, Zhaoyi You, Shijie Tang, Carl Kingsford, “Methods and Systems for Multi-Objective Codon Optimization of mRNA Sequences Using Reinforcement Learning”, patent application filed.

- Jiayi Li, Hong-Sheng Lai, Litian Liang, Shiyi Du, Shijie Tang, **Gün Kaynar**, Carl Kingsford, “Controllable Codon Design from Foundation Models via Activation Engineering”, patent application filed.

Academic Duty

Reviewer

- **NeurIPS 2026** First Workshop on Reliable Evaluation for Language Models
- **NeurIPS 2026** Workshop Who Verifies the Agents? Toward Reliable Agent Development
- **ACL ARR 2026** Association for Computational Linguistics Rolling Review
- **ICML 2026** Workshop on Failure Modes of Agentic AI at the 43rd International Conference on Machine Learning
- **RECOMB 2026** 30th Annual International Conference on Research in Computational Molecular Biology
- **ACM TCBB** IEEE/ACM Transactions on Computational Biology and Bioinformatics 2025
- **ISMB/ECCB 2025** 33nd Edition of The International Conference on Intelligent Systems for Molecular Biology
- **RECOMB 2025** 29th Annual International Conference on Research in Computational Molecular Biology
- **ACM TCBB** IEEE/ACM Transactions on Computational Biology and Bioinformatics 2024
- **ISMB 2024** 32nd Edition of The International Conference on Intelligent Systems for Molecular Biology
- **RECOMB 2024** 28th Annual International Conference on Research in Computational Molecular Biology
- **RECOMB 2023** 27th Annual International Conference on Research in Computational Molecular Biology
- **ACM BCB 2022** The 13th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics
- **ISBRA2022** 18th International Symposium on Bioinformatics Research and Applications
- **ACM TCBB** IEEE/ACM Transactions on Computational Biology and Bioinformatics 2022

Service

- **CMLH 2025** Fellowship Applicant Reviewer, Center for Machine Learning and Health, Carnegie Mellon University

Contributions

- **Humanity’s Last Exam** Contributed 2 questions in computational biology to the benchmark.

Organization

- **RECOMB 2023** 27th Annual International Conference on Research in Computational Molecular Biology
- **RECOMB-SEQ 2023** The 13th RECOMB Satellite Conference on Biological Sequence Analysis
- **RECOMB-CCB** The 15th RECOMB Satellite Workshop on Computational Cancer Biology
- **RECOMB-CG** The 20th RECOMB Satellite Conference on Comparative Genomics
- **RECOMB-Genetics 2023** The 11th RECOMB Satellite Workshop on Computational Methods in Genetics
- **BIO-Arch** Workshop on Hardware Acceleration of Bioinformatics Workloads

Experience

- 2024-Present **Graduate Research Assistant**, *School of Computer Science, Kingsford Lab*, Carnegie Mellon University, Pittsburgh, PA, US.
- 2026-Present **Anthropic-Arm-CMU**, *Model Hardware Standard (MHS) Research Preview*, Carnegie Mellon University, Pittsburgh, PA, US.
AI-agent orchestration of laboratory instruments for autonomous experimentation, including liquid handlers, plate readers, and robotic arms, using a shared specification for safe hardware operation.
- May 2026 **Arm-CMU Collaboration**, *Multi-Objective RNA Sequence Optimization*, Carnegie Mellon
Present University, Pittsburgh, PA, US.
Multi-objective RNA sequence optimization using dynamic programming, local search, and reinforcement learning.
- 2026-Present **CMU-Optum Research Partnership**, *with Prof. Giulia Fanti*, Carnegie Mellon University, Pittsburgh, PA, US.
Differentially private generation of temporal electronic health record data.
- Sep 2025 **Arm-CMU Collaboration**, *Programmable Cloud Laboratories*, Carnegie Mellon Univer-
Present sity, Pittsburgh, PA, US.
Simulation, orchestration, error handling, agentic supervision, and scheduling for programmable cloud laboratories.
- May 2025 **CMU-BioForge Research Partnership**, *mRNA Sequence Optimization*, Carnegie Mellon
Present University, Pittsburgh, PA, US.
Computational design and optimization of mRNA sequences with experimental production and validation by BioForge.
- 2024-Present **Graduate Teaching Assistant**, Carnegie Mellon University, Pittsburgh, PA, US.
◦ Automation of Scientific Research (Spring 2026)
◦ Algorithms and Advanced Data Structures (Fall 2025)
- 2021-2024 **Graduate Research Assistant**, *Computer Engineering Department, CICEKLAB, Sup-
ported by Scientific and Technological Research Council of Turkey*, Bilkent University, Ankara, Turkey.
- 2021-2024 **Graduate Teaching Assistant**, Bilkent University, Ankara, Turkey.
◦ CS473/CS573: Algorithms I (Spring 2024, Fall 2023, Spring 2023, Fall 2022)
◦ Introduction to Python and Artificial Intelligence (Summer 2023, Summer 2022)
◦ CS115: Introduction to Programming in Python (Spring 2022, Fall 2021)
- 2019 **Undergraduate Research Intern**, *CICEKLAB, under the supervision of Dr. A. Ercument Cicek*, Bilkent University, Ankara, Turkey.
Developed Nomind, a research project during the internship.
- 2018 **Undergraduate Research Intern**, *Center for Proteomics and Bioinformatics, under the supervision of Prof. Mark Chance*, Case Western Reserve University, Cleveland, OH, USA.
Developed Phosphorylator, a research project during the internship.

Training and Certificates

- Aug 2025 **Ph.D. Certificate in Ethics and Artificial Intelligence**, *Joint between SCS and Dietrich*
Present *College of Humanities and Social Sciences*, Carnegie Mellon University.
- Aug 2025 **HPC and Data Science Summer Institute**, *San Diego Supercomputer Center, UC San*
Diego, San Diego, CA, US.
Workshop on High Performance Computing

Software Systems (GitHub)

GitHub repositories for paper implementations and research projects:

-  seq2ribo (10+ ★)
-  TASER-AE
-  LYCEUM

-  ECOLE (30+ ★)
-  PiDeeL
-  Targeted Metabolomics